ECET 250E-Linear Circuit 1
Problemset 4-Chapter 4- Circuit Theorems
Solution

Name: _____ Marks: _____

Section 4.3Superposition

1. **Chapter 4, Question 11.** Use the superposition principle to find io and vo in the circuit of **Fig. 4.79**.

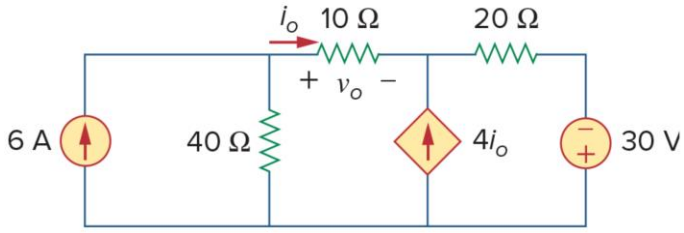
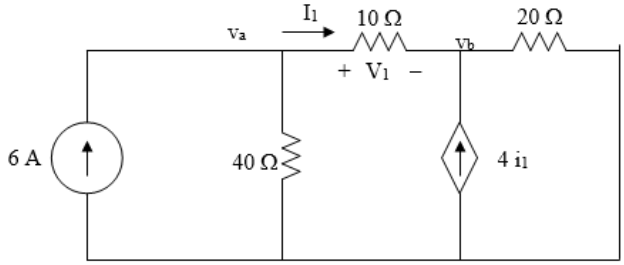


Figure 4.79 For **Prob. 4.11**.

Let $v_o = v_1 + v_2$, where v_1 and v_2 are due to the 6-A and 30-V sources respectively. To find $\mathbf{v_1}$, consider the circuit below.



At node a,

$$6 = \frac{v_a}{40} + \frac{v_a - v_b}{10} \longrightarrow 240 = 5v_a - 4v_b \qquad (1)$$

At node b,

$$-I_1 - 4I_1 + (v_b - 0)/20 = 0 \text{ or } v_b = 100I_1$$

But $i_1 = \frac{v_a - v_b}{10}$ which leads to $100(v_a - v_b)10 = v_b$ or $v_b = 0.9091v_a$ (2)

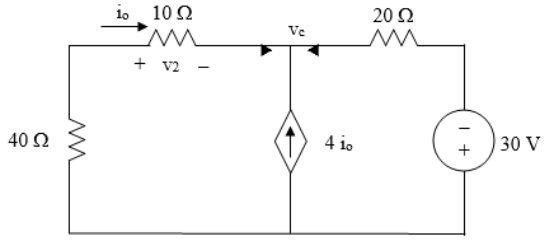
Substituting (2) into (1),

$$5v_a - 3.636v_a = 240 \text{ or } v_a = 175.95 \text{ and } v_b = 159.96$$

Reference: Fundamentals of Electric Circuits 7th edition, Charles Alexander and Matthew Sadiku, McGraw-Hill

However, $v_1 = v_a - v_b = 15.99$ V.

To find **v_2,** consider the circuit below



$$\frac{0-v_c}{50} + 4i_o + \frac{(-30-v_c)}{20} = 0$$

But $i_o = \dfrac{(0-v_c)}{50}$

$$-\frac{5v_c}{50} - \frac{(30+v_c)}{20} = 0 \longrightarrow v_c = -10 \text{ V}$$

$$i_2 = \frac{0-v_c}{50} = \frac{0+10}{50} = \frac{1}{5}$$

$$v_2 = 10i_2 = 2 \text{ V}$$

$v_o = v_1 + v_2 = 15.99 + 2 =$ **17.99 V** and $i_o = v_o/10 =$ **1.799 A**.

2. Chapter 4, Question 12. Determine Vo in the circuit of **Fig. 4.80** using the superposition principle.

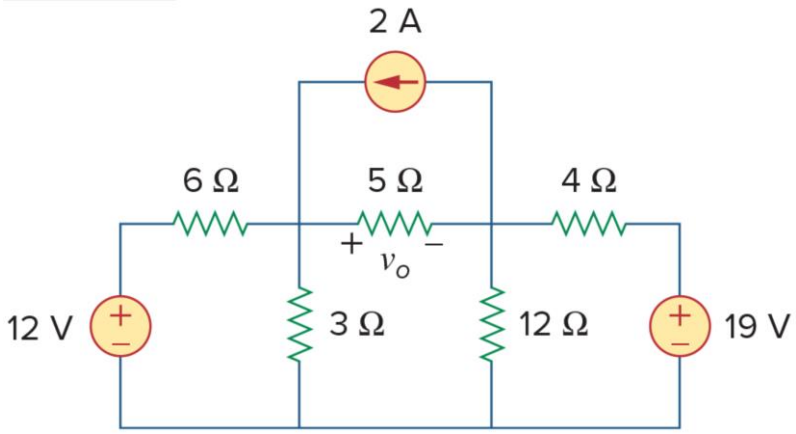
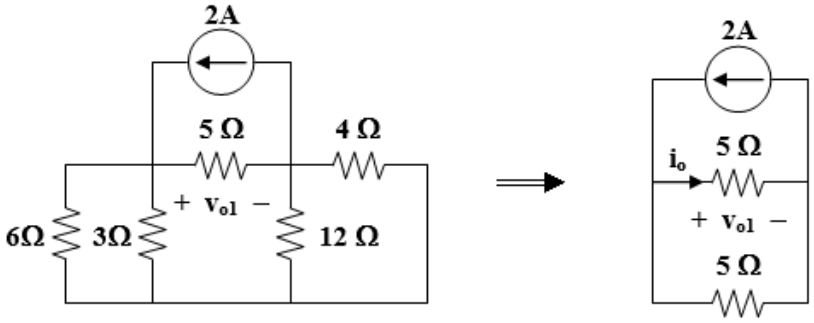


Figure 4.8 For **Prob. 4.12**.

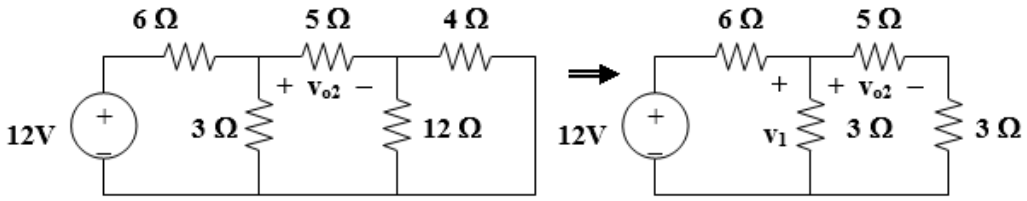
Let $v_o = v_{o1} + v_{o2} + v_{o3}$, where v_{o1}, v_{o2}, and v_{o3} are due to the 2-A, 12-V, and 19-V sources respectively. For v_{o1}, consider the circuit below.



$6||3 = 2$ ohms, $4||12 = 3$ ohms. Hence,

$$i_o = 2/2 = 1, \quad v_{o1} = 5io = 5 \text{ V}$$

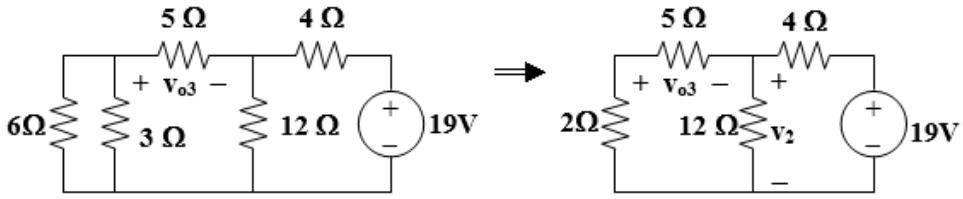
For v_{o2}, consider the circuit below.



$$3||8 = 24/11, \quad v_1 = [(24/11)/(6 + 24/11)]12 = 16/5$$

$$v_{o2} = (5/8)v_1 = (5/8)(16/5) = 2 \text{ V}$$

For v_{o3}, consider the circuit shown below.



$7||12 = (84/19)$ ohms, $v_2 = [(84/19)/ (4 + 84/19)]19 = 9.975$

$$v = (-5/7)\, v2 = -7.125$$

$$v_o = 5 + 2 - 7.125 = \mathbf{-125\ mV}$$

3. **Chapter 4, Question 16.** Given the circuit in **Fig. 4.84**, use superposition to obtain io.

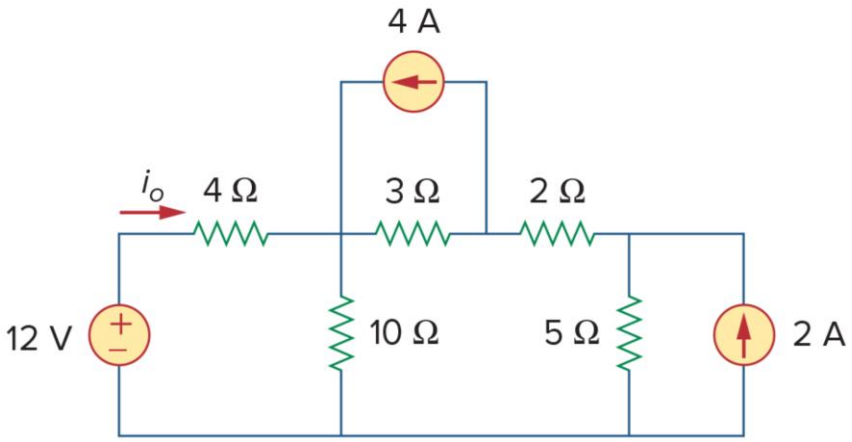
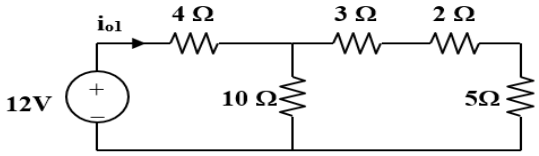


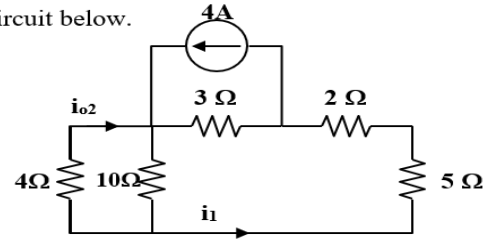
Figure 4.84 For **Prob. 4.16**.

Let $i_o = i_{o1} + i_{o2} + i_{o3}$, where i_{o1}, i_{o2}, and i_{o3} are due to the 12-V, 4-A, and 2-A sources. For i_{o1}, consider the circuit below.



$10||(3 + 2 + 5) = 5$ ohms, $i_{o1} = 12/(5 + 4) = (12/9)$ A

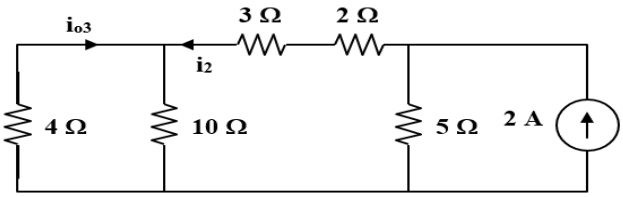
For i_{o2}, consider the circuit below.



$2 + 5 + 4||10 = 7 + 40/14 = 69/7$

$i_1 = [3/(3 + 69/7)]4 = 84/90$, $i_{o2} = [-10/(4 + 10)]i_1 = -6/9$

For i_{o3}, consider the circuit below.



$3 + 2 + 4||10 = 5 + 20/7 = 55/7$

$i_2 = [5/(5 + 55/7)]2 = 7/9$, $i_{o3} = [-10/(10 + 4)]i_2 = -5/9$

$i_o = (12/9) - (6/9) - (5/9) = 1/9 = \mathbf{111.11\ mA}$

4. **Chapter 4, Question 33.** Determine the Thevenin equivalent circuit, shown in **Fig. 4.101**, as seen by the 5-ohm resistor. Then calculate the current flowing through the 5-ohm resistor.

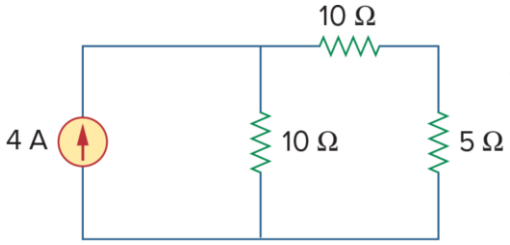
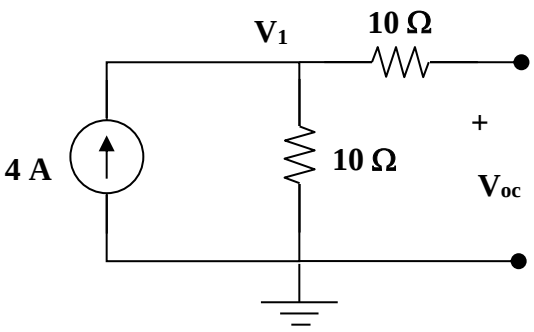
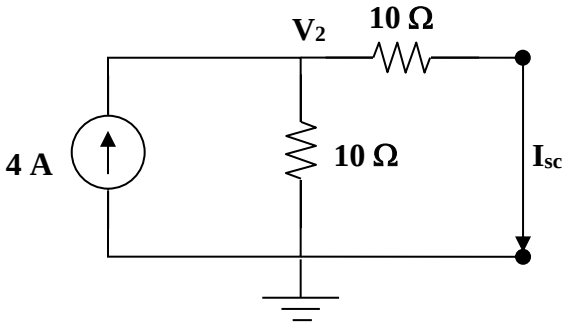


Figure 4.101 For **Prob. 4.33**.

Step 1. We need to find V_{oc} and I_{sc}. To do this, we will need two circuits, label the appropriate unknowns and solve for V_{oc}, I_{sc}, and then R_{eq} which is equal to V_{oc}/I_{sc}.



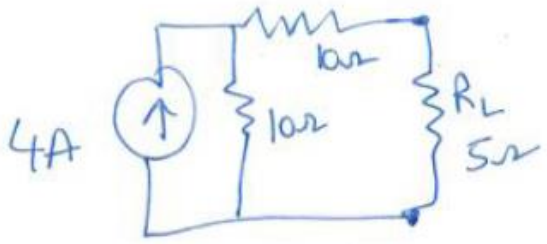


Note, in the first case $v_1 = V_{oc}$ and the nodal equation at 1 produces $-4+(V_1-0)/10 = 0$. In the second case, $I_{sc} = (V_2-0)/10$ where the nodal equation at 2 produces, $-4+[(V_2-0)/10]+[(V_2-0)/10] = 0$.

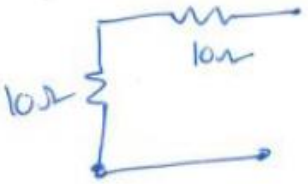
Step 2. $0.1V_1 = 4$ or $V_1 = \mathbf{40\ V} = V_{oc} = V_{Thev}$. Next, $(0.1+0.1)\ V_2 = 4$ or $0.2V_2 = 4$ or $V_2 = 20$ V. Thus, $I_{sc} = 20/10 = 2$ A. This leads to $R_{eq} = 40/2 = \mathbf{20\ \Omega}$. We can check our results by using source transformation. The 4-amp current source in parallel with the 10-ohm resistor can be replaced by a 40-volt voltage source in series with a 10-ohm resistor which in turn is in series with the other 10-ohm resistor yielding the same Thevenin equivalent circuit. Once the 5-ohm resistor is connected to the Thevenin equivalent circuit, we now have 40 V across 25 Ω producing a current of **1.6 A**.

Solutions Ps-4

1) 5ohms is the load Resistor

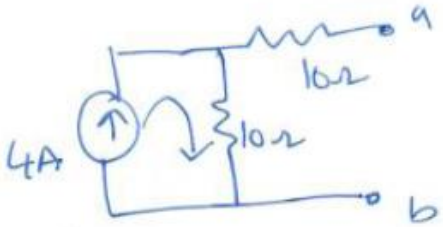


R_{th}



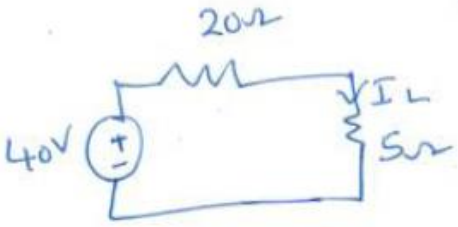
$\Rightarrow R_{th} = 20\Omega$

$V_{th} =$



$V_{ab} = V_{th} = V_{10\Omega} = 10 * 4$

$= 40V$



$$I_L = \frac{40}{25} = \frac{8}{5} = 1.6\ A$$

5. **Chapter 4, Question 37** Find the Norton equivalent with respect to terminals *a-b* in the circuit shown in **Fig. 4.104**.

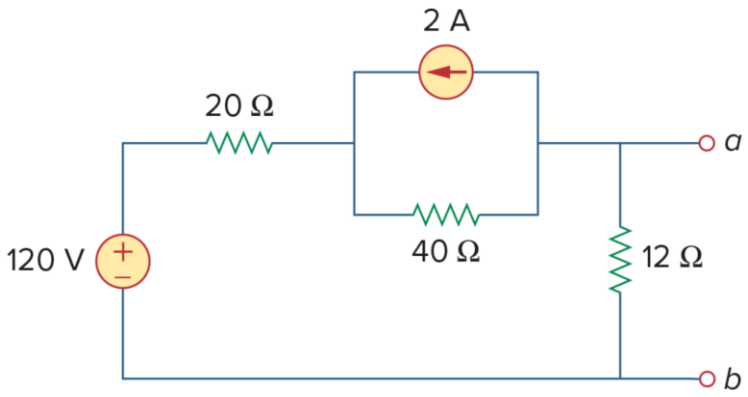
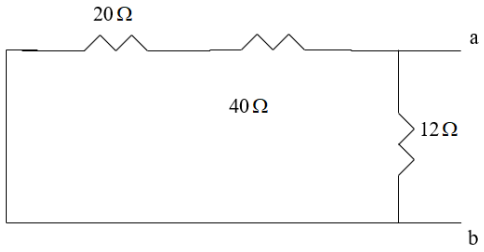


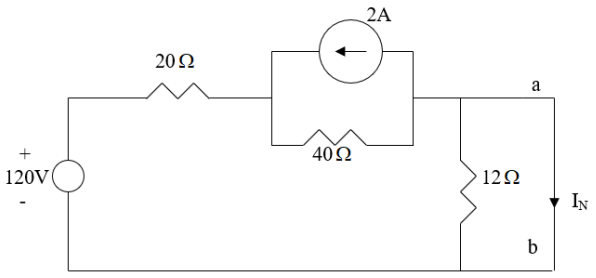
Figure 4.104 For **Prob. 4.37**.

R_N is found from the circuit below.

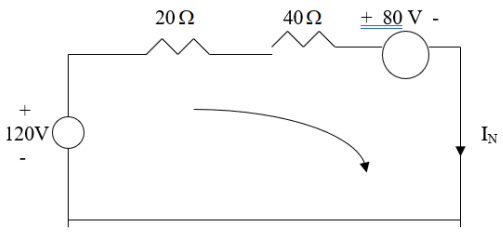


$R_N = 12 // (20 + 40) = \mathbf{10\ \Omega}$

I_N is found from the circuit below.

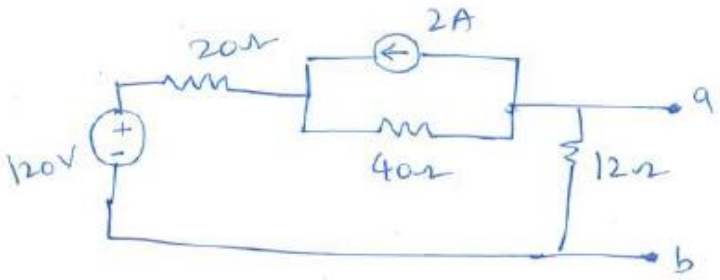


Applying source transformation to the current source yields the circuit below.

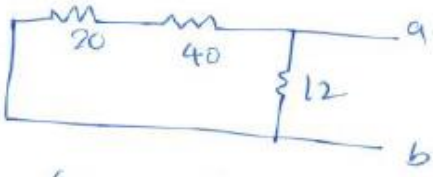


Applying KVL to the loop yields

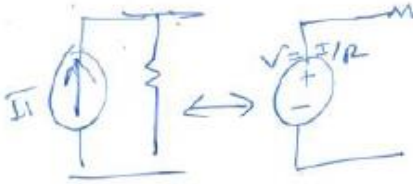
$-120 + 80 + 60 I_N = 0 \longrightarrow I_N = 40/60 =$ **666.7 mA**.



R_{th}

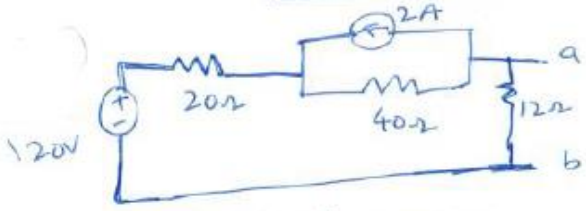


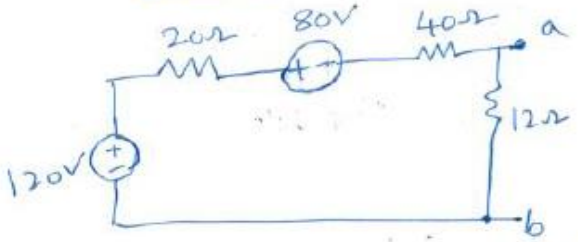
$(20+40) \,||\, 12$



$$\frac{60 \times 12}{60+12} = \frac{10}{\frac{60 \times 12}{72}}$$

$$R_{th} = 10\,\Omega$$





$$-120 + 20i + 40i + 12i + 80 = 0$$

$$-40 = -72i$$

$$i = 40/72 = 0.56$$

$$\Rightarrow i = 0.56$$

$$V_{th} = V_{12\Omega} = (0.56) \times 12 = 6.67\,V$$

6. Chapter 4, Question 36. Solve for the current i in the circuit of **Fig. 4.103** using Thevenin's theorem. (*Hint:* Find the Thevenin equivalent seen by the 40Ω resistor.)

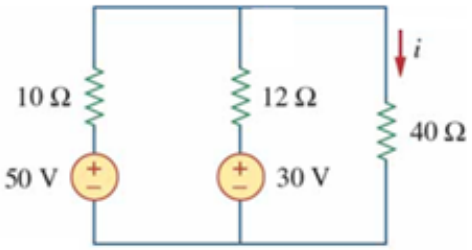
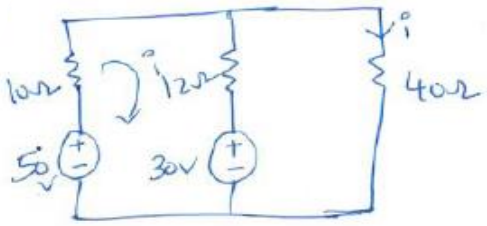
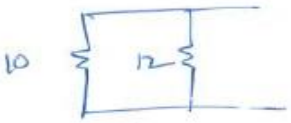


Figure 4.103 For **Prob. 4.36**.



R_{th}



$$\frac{10\times12}{10+12} = \frac{10\times12}{22} = \frac{120}{22} = \frac{60}{11} = 5.45\ \Omega$$

V_{th} =

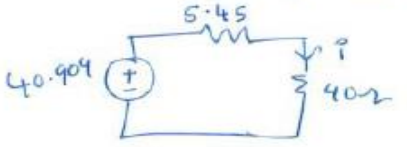
$$-50 + 10i + 12i + 30 = 0$$

$$22i = 20$$

$$i = \frac{10}{11} = 0.909\ A$$

$$V_{th} = 12 * i + 30$$

$$= 40.909\ V$$



$$\Rightarrow \quad i = \frac{40.909}{5.45+40}$$

$$i = 0.9\ A$$

Reference: Fundamentals of Electric Circuits 7th edition, Charles Alexander and Matthew Sadiku, McGraw-Hill

7. **Chapter 4, Question 38.** Apply Thevenin's theorem to find Vo in the circuit of **Fig. 4.105**.

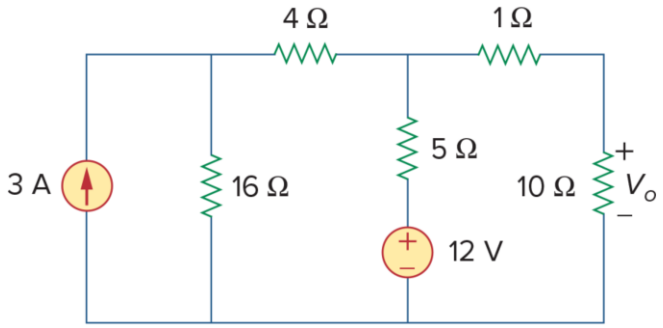
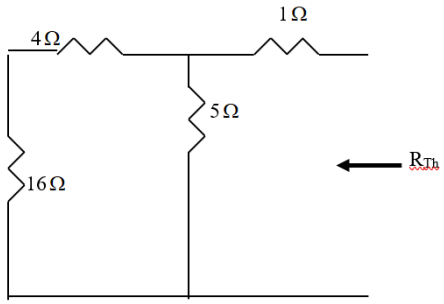


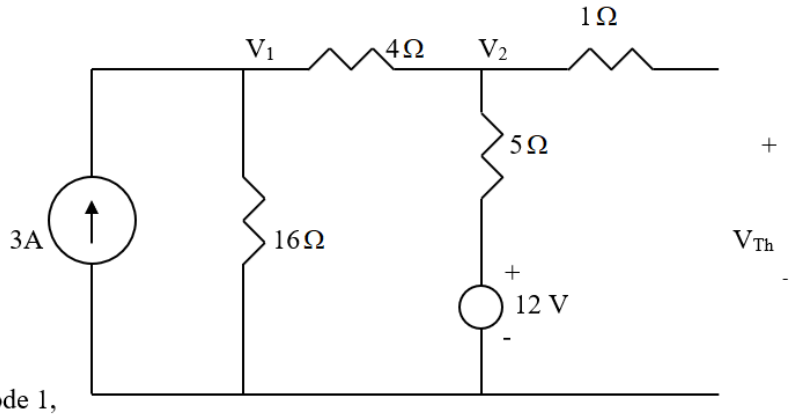
Figure 4.105 For **Prob. 4.38**.

We find Thevenin equivalent at the terminals of the 10-ohm resistor. For R_{Th}, consider the circuit below.



$$R_{Th} = 1 + 5 // (4 + 16) = 1 + 4 = 5\Omega$$

For V_{Th}, consider the circuit below.



At node 1,

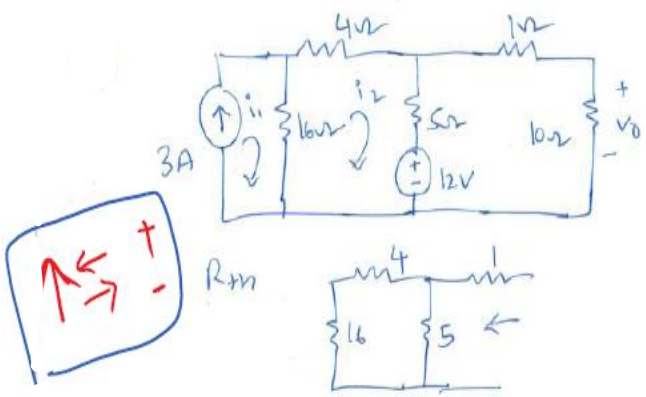
$$3 = \frac{V_1}{16} + \frac{V_1 - V_2}{4} \longrightarrow 48 = 5V_1 - 4V_2 \qquad (1)$$

At node 2,

$$\frac{V_1 - V_2}{4} + \frac{12 - V_2}{5} = 0 \longrightarrow 48 = -5V_1 + 9V_2 \qquad (2)$$

Solving (1) and (2) leads to

$$V_{Th} = V_2 = 19.2$$



$$R_{th} \quad \frac{20 \times 5}{25} = 4$$

$$R_{th} = 4 + 1 = 5\Omega$$

$i_1 = 3A$ from loop 1

V_{th}

$$16(i_2 - i_1) + 4i_2 + 5i_2 + 12 = 0$$

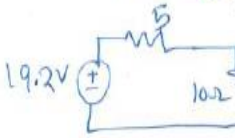
$$i_1 = 3A$$

$$-48 + 16i_2 + 4i_2 + 5i_2 + 12 = 0$$

$$25i_2 = 36$$

$$i_2 = \frac{36}{25}$$

$$V_{th} = +5i_2 + 12 = 19.2\ V$$

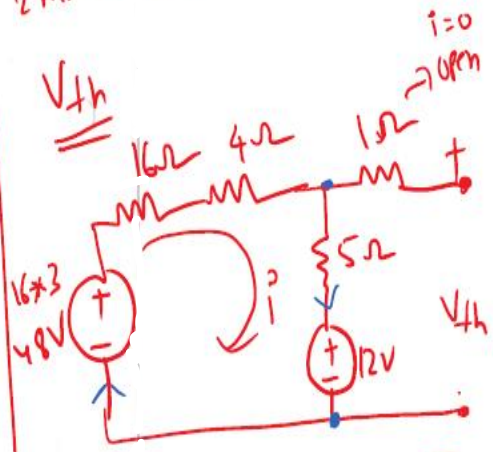


$$i = \frac{19.2}{5+10} = 1.28$$

$$V_0 = 12.8\ V$$

2nd method.

V_{th}



$$V_{th} = 5i + 12V$$

KVL

$$-48 + 16i + 4i + 5i + 12 = 0$$

$$i = \frac{36}{25}$$

$$V_{th} = 5i + 12$$

$$= 19.2\ V$$

Reference: Fundamentals of Electric Circuits 7th edition, Charles Alexander and Matthew Sadiku, McGraw-Hill

8. Chapter 4, Question 39. Obtain the Thevenin equivalent at terminals *a-b* of the circuit shown in **Fig. 4.106**.

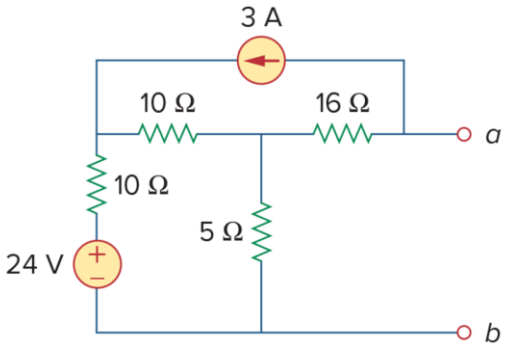
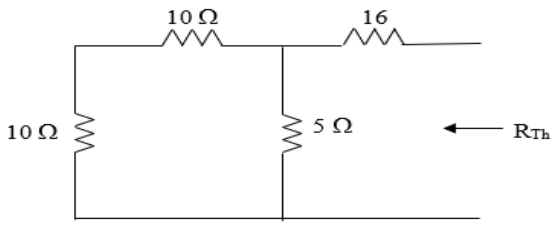


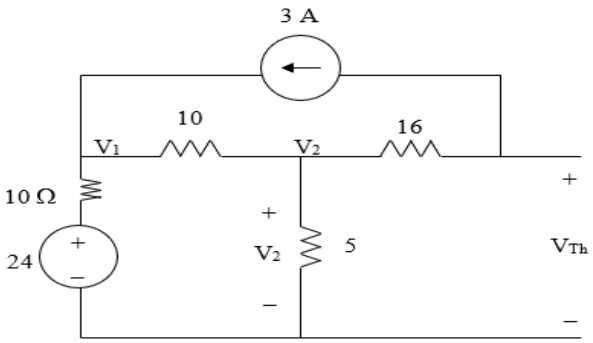
Figure 4.106 For **Prob. 4.39**.

We obtain R_{Th} using the circuit below.



$V_{Thev} = 16 + (20||5) = 16 + (20x5)/(20+5) = \mathbf{20\ \Omega}$

To find V_{Th}, we use the circuit below.



At node 1,

$$\frac{24 - V_1}{10} + 3 = \frac{V_1 - V_2}{10} \longrightarrow 54 = 2V_1 - V_2 \qquad (1)$$

At node 2,

$$\frac{V_1 - V_2}{10} = 3 + \frac{V_2}{5} \longrightarrow 60 = 2V_1 - 6V_2 \qquad (2)$$

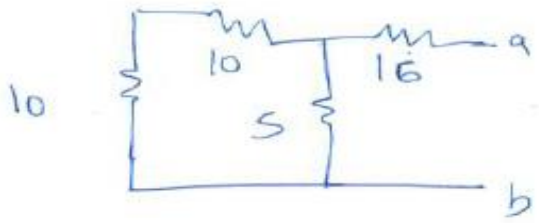
Subtracting (1) from (2) gives

$$6 = -5V_2 \text{ or } V_2 = -1.2 \text{ V}$$

But

$$-V_2 + 16x3 + V_{Thev} = 0 \text{ or } V_{Thev} = -(48 + 1.2) = \mathbf{-49.2\ V}$$

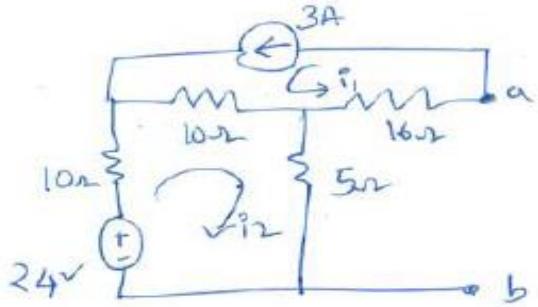
R_{th}



$(20 \| 5) + 16$

$\frac{20 \times 5}{25} + 16 = 20\Omega$

V_{th}



$i_1 = 3A$

$-24 + 10i_2 + 10(i_2 + i_1) + 5i_2 = 0$

$25i_2 + 10i_1 = 24$

$25i_2 + 10(3) = 24$

$i_2 = -6/25 = -0.24$

$V_{5\Omega} = 5i_2 = -1.2\,V$

Reference: Fundamentals of Electric Circuits 7th edition, Charles Alexander and Matthew Sadiku, McGraw-Hill

9. **Chapter 4, Question 41.** Find the Thevenin and Norton equivalents at terminals *a-b* of the circuit shown in **Fig. 4.108**.

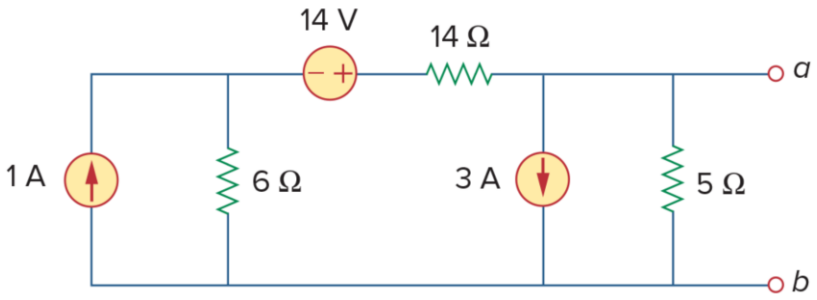
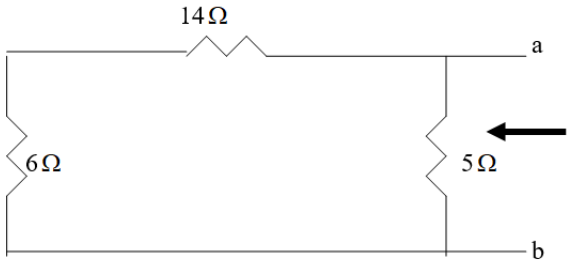


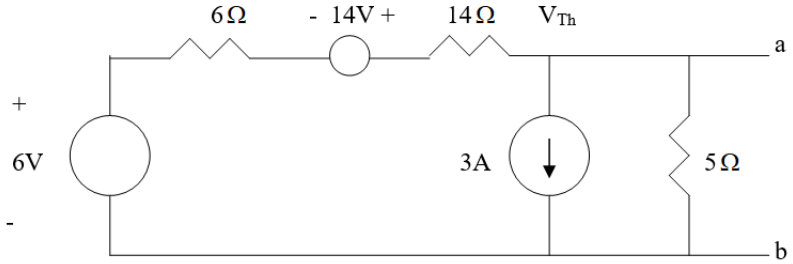
Figure 4.108 For **Prob. 4.41**.

To find R_{Th}, consider the circuit below



$$R_{Th} = 5 // (14 + 6) = 4\Omega = R_N$$

Applying source transformation to the 1-A current source, we obtain the circuit below.



At node a,

$$\frac{14 + 6 - V_{Th}}{6 + 14} = 3 + \frac{V_{Th}}{5} \quad \longrightarrow \quad V_{Th} = -8\,\text{V}$$

$$I_N = \frac{V_{Th}}{R_{Th}} = (-8)/4 = -2\,\text{A}$$

Thus,

$$\boldsymbol{R_{Th} = R_N = 4\Omega, \quad V_{Th} = -8\text{V}, \quad I_N = -2\,\text{A}}$$

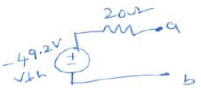
Reference: Fundamentals of Electric Circuits 7th edition, Charles Alexander and Matthew Sadiku, McGraw-Hill

$V_{ab} = V_{th} =$

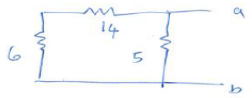
$-5i_2 + 16 \times i_1 + V_{ab} = 0$

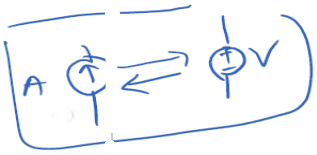
$+1.2 + 48 = -V_{ab}$

$V_{th} = V_{ab} = -49.2\ V$

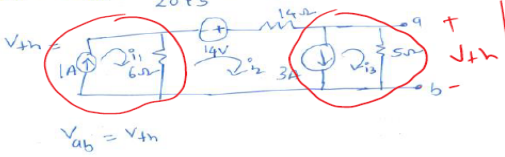


R_{th}



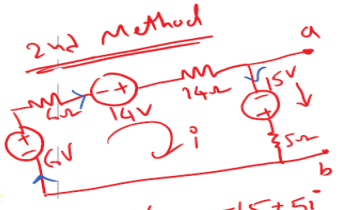


$R_N = R_{th} = \frac{20 \times 5}{20 + 5} = 4\Omega$



$V_{ab} = V_{th}$

2nd Method



$V_{ab} = V_{th} = -15 + 5i$

$-6 + 6i + 14i + 5i - 15 - 14 = 0$

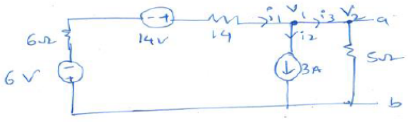
$i = 35/25 = 7/5$

$V_{th} = -15 + 5(\frac{7}{5})$

$= -15 + 7$

$= -8V$

Convert c.s to v.s



$V_{ab} = V_2 = V_{th} = V_1$

V_1 : $i_1 = i_2 + i_3$

$\frac{(6 - (-14)) - V_1}{6 + 14} = 3 + \frac{V_2}{5}$

$\frac{20 - V_1}{20} = 3 + \frac{V_2}{5}$ $\quad V_1 = V_2$

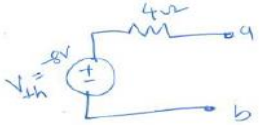
$\frac{20 - V_2}{20_4} = \frac{15 + V_2}{5_1}$

$20 - V_2 = 60 + 4V_2$

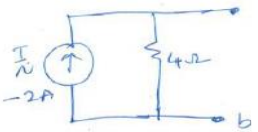
$5V_2 = -40$

$V_2 = -8\ V$

$V_{th} = -8V \Rightarrow I_N = \frac{V_{th}}{R_N} = \frac{-8}{4} = -2A$



Thevenin's



Norton's

Reference: Fundamentals of Electric Circuits 7th edition, Charles Alexander and Matthew Sadiku, McGraw-Hill

Section 4.8Maximum Power Transfer

10. Chapter 4, Question 66. Find the maximum power that can be delivered to the resistor *R* in the circuit of **Fig. 4.132**.

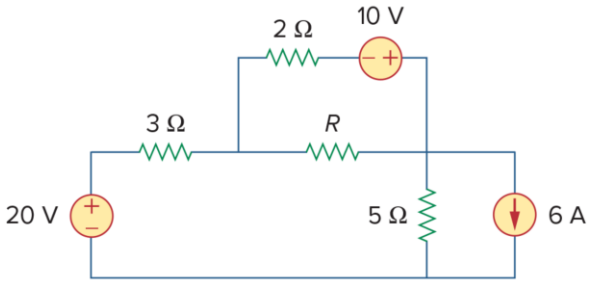
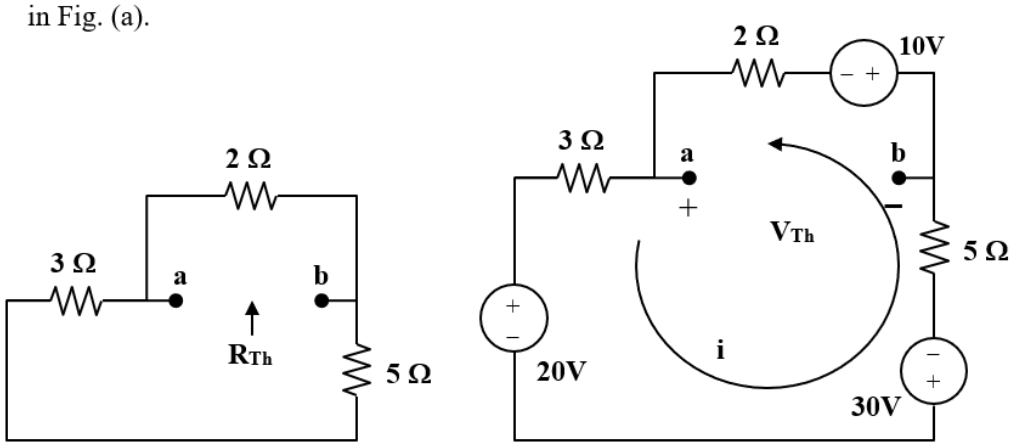


Figure 4.132 For **Prob. 4.66**.

We first find the Thevenin equivalent at terminals a and b. We find R_{Th} using the circuit in Fig. (a).



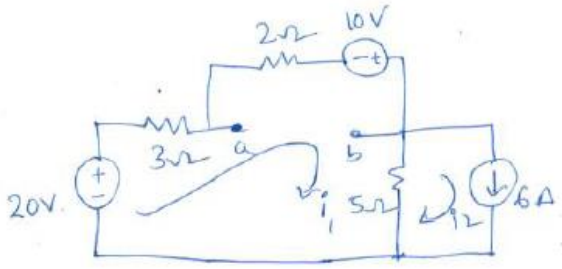
$$R_{Th} = 2||(3+5) = 2||8 = \mathbf{1.6\ ohms}$$

By performing source transformation on the given circuit, we obatin the circuit in (b). We now use this to find V_{Th}.

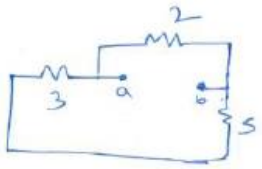
$$10i + 30 + 20 + 10 = 0, \text{ or } i = -6$$

$$V_{Th} + 10 + 2i = 0, \text{ or } V_{Th} = 2\ V$$

$$p = V_{Th}^2/(4R_{Th}) = (2)^2/[4(1.6)] = \mathbf{625\ m\ watts}$$



R_{th} =



$$R_{ab} = R_{th} = (3+5) \parallel 2$$

$$= 8 \parallel 2$$

$$= \frac{2 \times 8}{10} = 1.6\ \Omega$$

$i_2 = 6A$

V_{th} =

$$-20 + 3i_1 + 2i_1 - 10 + 5(i_1 - i_2) = 0$$

$$10i_1 - 30 - 5i_2 = 0$$

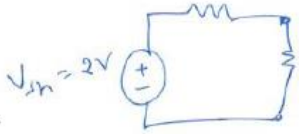
$$i_1 = \frac{+30 + 5(6)}{10} = 6A$$

$$V_{ab} = V_{th} = 2i_1 - 10$$

$$= 2(6) - 10$$

$$\boxed{V_{ab} = 2V}$$

$R_{th} = 1.6\Omega$



$R = R_{th} = 1.6\Omega$ to get max power

$$P_{max} = \frac{(V_{th})^2}{4R_{th}} = \frac{2^2}{4 \times 1.6} = 625\ mW$$

Reference: Fundamentals of Electric Circuits 7th edition, Charles Alexander and Matthew Sadiku, McGraw-Hill